Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following transformation $\frac{\mathrm{d}u}{\mathrm{d}x} + uP(x) = Q(x)$

(A) $u = \log z$

(B) $u = e^z$

(C) $u = (\log z)^{-1}$

(D) $u = (\log z)^2$

Q.2

The solution of $\frac{x\,\mathrm{d}y}{x^2+y^2} = \left(\frac{y}{x^2+y^2} - 1\right)\mathrm{d}x$ is

(A) $y = x\cot(c - x)$

(B) $\cos^{-1}(y/x) = -x + c$

(C) $y = x\tan(c - x)$

(D) $y^2/x^2 = x\tan(c - x)$

Q.3

The solution of $y^5 x + y - x\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ is

(A) $x^4/4 + (1/5)(x/y)^5 = C$

(B) $x^5/5 + (1/4)(x/y)^4 = C$

(C) $(x/y)^5 + x^4/4 = C$

(D) $(xy)^4 + x^5/5 = C$

Q.4

An equation of the curve satisfying $x\,\mathrm{d}y - y\,\mathrm{d}x = \sqrt{x^2 - y^2}\,\mathrm{d}x$ and $y(1) = 0$ is

(A) $y = x^2\log|\sin x|$

(B) $y = x\sin(\log|x|)$

(C) $y^2 = x(x-1)^2$

(D) $y = 2x^2(x-1)$

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Q.5

The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is

(A) $y = \log x + x$

(B) $y = x \log x + x^2$

(C) $y = xe^{(x-1)}$

(D) $y = x \log x + x$

Q.6

At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is

(A) 3500

(B) 4500

(C) 2500

(D) 3000

Q.7

The slope of normal at any point *P* of a curve (lying in the first quadrant) is reciprocal of twice the product of the abscissa and the ordinate of point *P*. Then, the equation of the curve is (where, *c* is an arbitrary constant)

(A) $y^2 = x + c$

(B) $y = ce^{-x^2}$

(C) $y = ce^{-x}$

(D) $y^2 = \ln x + c$

Q.8

A function $y = f(x)$ satisfies the condition $f'(x) \sin x + f(x) \cos x = 1, f(x)$ being bounded when $x \to 0$

If $l = \int_0^{\pi/2} f(x)dx$, then

(A) $\frac{\pi}{2} < l < \frac{\pi^2}{4}$

(B) $\frac{\pi}{4} < l < \frac{\pi^2}{2}$

(C) $1 < l < \frac{\pi}{2}$

(D) $0 < l < 1$

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Q.9

A function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} - y = \cos x - \sin x$ with initial condition that y is bounded when $x \to \infty$. The area enclosed by $y = f(x), y = \cos x$ and the y -axis is

(A) $\sqrt{2} - 1$

(B) $\sqrt{2}$

(C) 1

(D) $\frac{1}{\sqrt{2}}$

Q.10

The differential equation of the family of circles with fixed radius 5 units and centre on the line $y = 2$ is

(A) $(y-2)y'^2 = 25 - (y-2)^2$

(B) $(y-2)^2y'^2 = 25 - (y-2)^2$

(C) $(x-2)^2y'^2 = 25 - (y-2)^2$

(D) $(x-2)y'^2 = 25 - (y-2)^2$

Q.11

Which of the following does not represent the orthogonal trajectory of the system of curves $\left(\frac{dy}{dx}\right)^2 = \frac{a}{x}$

(A) $9a(y+c)^2 = 4x^3$

(B) $y + c = \frac{-2}{3\sqrt{a}}x^{3/2}$

(C) $y + c = \frac{2}{3\sqrt{a}}x^{3/2}$

(D) All are orthogonal trajectories

Q.12

The equation of the curve passing through the point $\left(0, \frac{\pi}{4}\right)$ whose differential equation is $\sin x \cos y dx + \cos x \sin y dy = 0$, is

(A) $\sec x \sec y = \sqrt{2}$

(B) $\cos x \cos y = \sqrt{2}$

(C) $\sec x = \sqrt{2}\cos y$

(D) $\cos \mathrm{y} = \sqrt{2}\sec \mathrm{y}$

Q.13

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Themarginal costofmanufacturing a certain item is given by $c'(x) = \frac{dc}{dx} = 2 + 0.15x$. Thetotal cost function $\mathrm{c}(\mathrm{x})$, is

(A) $0.075\mathrm{x}^2 + 2\mathrm{x} + 100$

(B) $0.15x^2 + 3x + 30$

(C) $x^2 + 100.075x + 100$

(D) None of these It is given that $c(0) = 100$

Q.14

The population of a country doubles in 40 years. Assuming that the rate of increase is proportional to the number of inhabitants, the number of years in which it would treble itself is

(A) 80 years

(B) $80\frac{\log 2}{\log 3}$ years

(C) $40\frac{\log 3}{\log 2}$ years

(D) $40 \log 2 \log 3$ years

Q.15

Under which one of the following conditions does the solution of $\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\mathrm{ax}+\mathrm{b}}{\mathrm{cy}+\mathrm{d}}$ represent a parabola?

(A) $\mathrm{a} = 0, \mathrm{c} = 0$

(B) $a = 1, b = 2, c \neq 0$

(C) $\mathrm{a} = 0, \mathrm{c} \neq 0, \ \mathrm{b} \neq 0$

(D) $a = 1, c = 1$

Q.16

The differential equation of all circles passing through the origin and having their centres on the x -axis is

(A) $y^2 = x^2 + 2xy\frac{dy}{dx}$

(B) $y^2 = x^2 - 2xy\frac{dy}{dx}$

(C) $x^2 = y^2 + xy\frac{dy}{dx}$

(D) $x^2 = y^2 + 3xy\frac{dy}{dr}$

Q.17

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If the gradient of the tangent at any point (x, y) of a curve which passes through the point $\left(1, \frac{\pi}{4}\right)$ is $\left\{\frac{y}{x}-\sin^2\left(\frac{y}{x}\right)\right\}$, then equation of the curve is

(A) $y=\cot^{-1}\left(\log_e x\right)$

(B) $y=\cot^{-1}\left(\log_e \frac{x}{e}\right)$

(C) $y=x\cot^{-1}\left(\log_e ex\right)$

(D) $y=\cot^{-1}\left(\log_e \frac{e}{x}\right)$

Q.18

The differential equation
$\left(1+y^2\right)xdx-\left(1+x^2\right)ydy=0$
represents a family of :

(A) ellipses of constant eccentricity

(B) ellipses of variable eccentricity

(C) hyperbolas of constant eccentricity

(D) hyperbolas of variable eccentricity

Q.19

The solution of the differential equation
$\left\{1+x\sqrt{\left(x^2+y^2\right)}\right\}dx+\left\{\sqrt{\left(x^2+y^2\right)}-1\right\}ydy=0$
is

(A) $x^2+\frac{y^2}{2}+\frac{1}{3}\left(x^2+y^2\right)^{3/2}=C$

(B) $x-\frac{y^2}{3}+\frac{1}{2}\left(x^2+y^2\right)^{1/2}=C$

(C) $x-\frac{y^2}{2}+\frac{1}{3}\left(x^2+y^2\right)^{3/2}=C$

(D) None of these

Q.20

If $\mathrm{f}(x)$ and $\mathrm{g}(x)$ are two solution of the differential equation
$a^2\frac{d^2y}{dx^2}+x^2\frac{dy}{dx}+y=e^x,$ then $\mathrm{f}(x)-\mathrm{g}(x)$ is the solution of

(A) $a^2\frac{d^2y}{dx^2}+\frac{dy}{dx}+y=e^x$

(B) $a^2\frac{d^2y}{dx^2}+y=e^x$

(C) $a^2\frac{d^2y}{dx^2}+x^2\frac{dy}{dx}+y=0$

(D) $a^2\frac{d^2y}{dx^2}-y=e^x$

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Q.1 Answer:

$u = (\log z)^{-1}$

Solution:

Dividing the given equation by $z(\log z)^2$, we have

$\frac{1}{z(\log z)^2}\frac{\mathrm{d}z}{\mathrm{d}x} + \frac{1}{\log z}\frac{1}{x} = \frac{1}{x^2}$

Putting $\frac{1}{\log z} = u$, we have $\frac{\mathrm{d}u}{\mathrm{d}x} = -(\log z)^{-2}\frac{1}{z}\frac{\mathrm{d}z}{\mathrm{d}x}$

So (i) can be written as

$-\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{u}{x} = \frac{1}{x^2} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} - \frac{u}{x} = \frac{-1}{x^2}$

which is in the required form with $P(x) = -1/x$ and $Q(x) = -1/x^2$

Q.2 Answer:

$y = x\tan(c - x)$

Solution:

The given equation can be written as

$\frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{x^2+y^2} = -\mathrm{d}x \Rightarrow \frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{x^2} \times \frac{1}{1+y^2/x^2} = -\mathrm{d}x$

$\Rightarrow \quad \frac{1}{1+y^2/x^2}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{x}\right) = -\mathrm{d}x$. **Integrating we have**

$\tan^{-1}(y/x) = -x + c$

$\Rightarrow \quad y = x\tan(c - x).$

Q.3 Answer:

$x^5/5 + (1/4)(x/y)^4 = C$

Solution:

The given differential equation can be written as $y^5 x\,\mathrm{d}x + y\,\mathrm{d}x - x\,\mathrm{d}y = 0$. Multiplying by x^3/y^5, we have

$x^4\,\mathrm{d}x + \frac{x^3}{y^3}\left(\frac{y\,\mathrm{d}x - x\,\mathrm{d}y}{y^2}\right) = 0$

Integrating, we get $x^5/5 + (1/4)(x/y)^4 = C$,

since

$\frac{x^3}{y^3}\left(\frac{y\,\mathrm{d}x - x\,\mathrm{d}y}{y^2}\right) = u^3\,\mathrm{d}u \quad$ where $u = \frac{x}{y}$

Q.4 Answer:

$y = x\sin(\log|x|)$

Solution:

The equation can be written as

$x^2\frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{x^2} = x\sqrt{1-(y/x)^2}\,\mathrm{d}x$

$\Rightarrow \quad \frac{\mathrm{d}(y/x)}{\sqrt{1-(y/x)^2}} = \frac{\mathrm{d}x}{x} \quad \Rightarrow \quad \sin^{-1} y/x = \log|x| + \text{ Const}$

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Since $y(1) = 0$ so Const $= 0$. Hence $y = x\sin(\log|x|)$. Note that one can also solve the given equation as a homogenous equation also, i.e. by putting $y = vx$

Q.5 Answer:

$y = x\log x + x$

Solution:

$xdy = xdx + ydx$

$\frac{xdy - ydx}{x^2} = \frac{dx}{x}$

$\Rightarrow \quad d(y/x) = d(\log x) \Rightarrow y = x(\log x + C)$

Putting $x = 1$, we get $C = 1$, so

$y = x(\log x + 1)$

Q.6 Answer:

3500

Solution:

The correct option is 1 $3500\frac{\mathrm{dP}}{\mathrm{dx}} = 100 - 12\sqrt{\mathrm{x}}$

$\Rightarrow \int \mathrm{dP} = \int (100 - 12\sqrt{\mathrm{x}})\mathrm{dx}$

$\Rightarrow \mathrm{P} = 100\mathrm{x} - 8\mathrm{x}^{3/2} + \mathrm{C}$

Now, $\mathbf{x = 0}$

Q.7 Answer:

$y = ce^{-x^2}$

Solution:

Slope of normal at $P = \frac{-1}{\frac{dy}{dx}_P}$

$\Rightarrow \frac{1}{2xy} = \frac{-1}{\frac{dy}{dx}}$

or $\frac{dy}{dx} = -2xy$

$\Rightarrow \int \frac{dy}{y} = \int -2xdx$

$\Rightarrow \ln y = -x^2 + k$

$\Rightarrow y = e^{-x^2 + k} = ce^{-x^2}$

Q.8 Answer:

$\frac{\pi}{2} < l < \frac{\pi^2}{4}$

Solution:

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$\sin x \frac{dy}{dx} + y \cos x = 1$

$\frac{dy}{dx} + y \cot x = \operatorname{cosec} x$

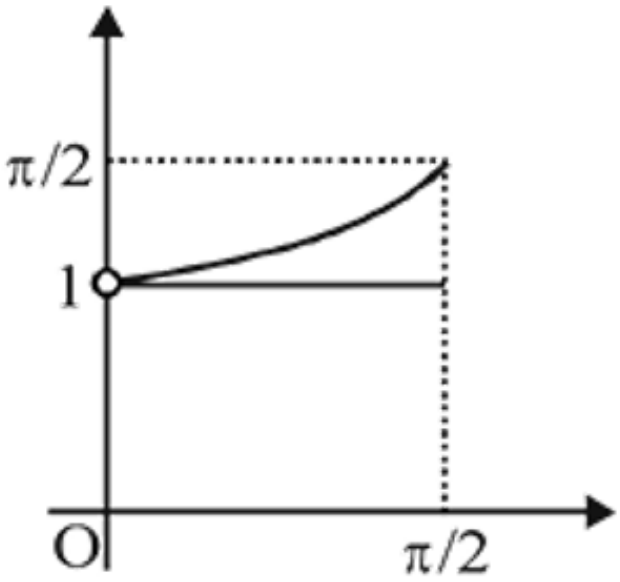
$\text{IF} = e^{\int \cot x dx} = e^{\ln(\sin x)} = \sin x$

$y \sin x = \int \operatorname{cosec} x \cdot \sin x dx = x + C$

If $x = 0, y$ is finite $\quad \therefore C = 0$

$y = x(\operatorname{cosec} x) = \frac{x}{\sin x}$

Now, $l < \frac{\pi^2}{4}$ and $l > \frac{\pi}{2}$ Hence, $\frac{\pi}{2} < l < \frac{\pi^2}{4}$



Q.9 Answer:

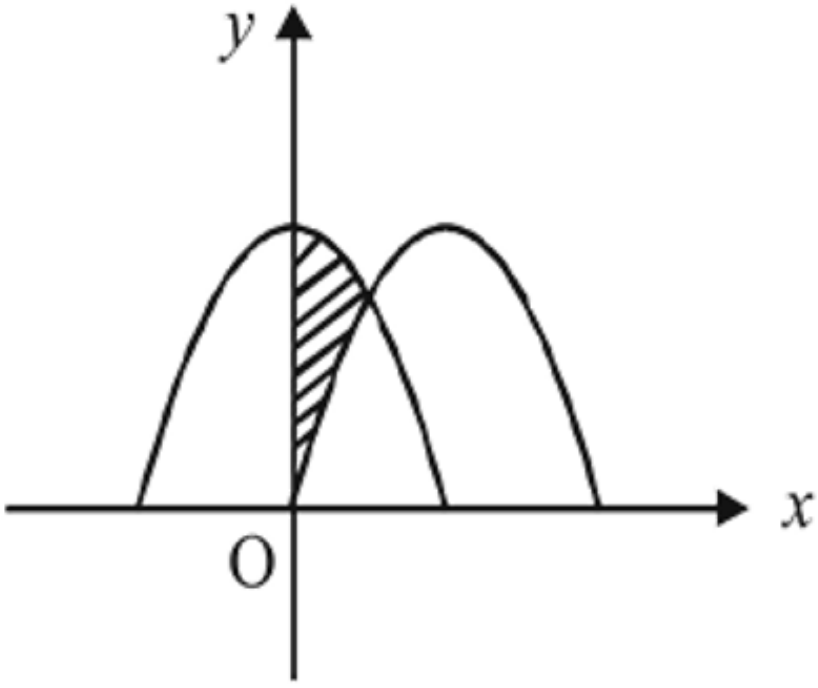
$\sqrt{2} - 1$

Solution:

$\text{IF} = e^{-x}$

$\therefore \quad ye^{-x} = \int e^{-x}(\cos x - \sin x)dx$

Put $-x = t$



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$= -\int e^t(\cos t + \sin t)dt = -e^t \sin t + c$

$ye^{-x} = e^{-x} \sin x + c$

Since, y is bounded when $x \to \infty \Rightarrow c = 0 \therefore \quad y = \sin x$

Area $= \int_0^{\pi/4}(\cos x - \sin x)dx = \sqrt{2} - 1$

Q.10 Answer:

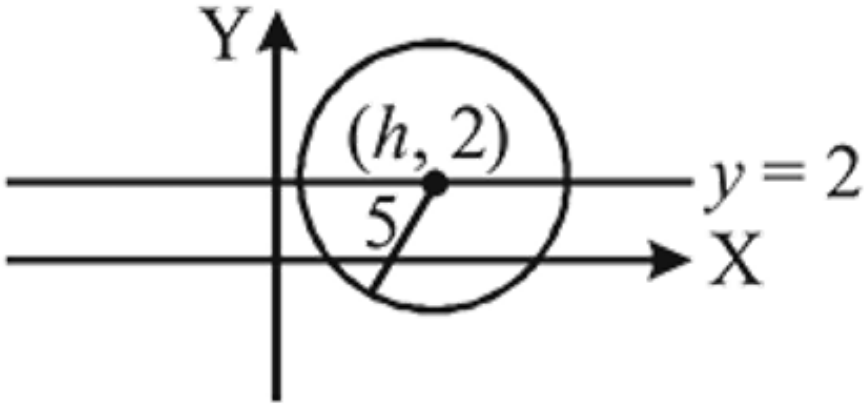
$(y-2)^2 y'^2 = 25 - (y-2)^2$

Solution:

$\text{Circle}(x-h)^2 + (y-2)^2 = 25$

$\Rightarrow 2(x-h) + 2(y-2)y' = 0 \quad \Rightarrow \quad 2(y-2)y'$

$= -2(x-h)$



$\Rightarrow \quad (y-2)^2 y'^2 = (x-h)^2 \Rightarrow (y-2)^2 y'^2 = 25 - (y-2)^2$

Q.11 Answer:

All are orthogonal trajectories

Solution:

The family of curves which are orthogonal (i.e. intersect at right angles) to a given system of curves is obtained by substituting $-\frac{dx}{dy}$ for $\frac{dy}{dx}$ in the differential equation of the given system. The given differential equation is $\left(\frac{dy}{dx}\right)^2 = \frac{a}{x}$

Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get

$\left(\frac{dx}{dy}\right)^2 = \frac{a}{x} \Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{x}{a} \Rightarrow \frac{dy}{dx} = \pm\sqrt{\frac{x}{a}}$

Integrating we get, $y + c = \pm\frac{2}{3\sqrt{a}}x^{3/2} \quad \ldots.$ (i)

$\Rightarrow (y+c)^2 = \frac{4}{9a}x^3 \Rightarrow 9a(y+c)^2 = 4x^3 \ldots$

..(ii) From (i) and (ii) all of the first three given options represent required equations.

Q.12 Answer:

$\sec x \sec y = \sqrt{2}$

Solution:

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The given differential equation is $\sin x \cos y dx + \cos x \sin y dy = 0$
dividing by $\cos x \cos y \Rightarrow \frac{\sin x}{\cos x} dx + \frac{\sin y}{\cos y} dy = 0$
Integrating, $\int \tan x dx + \int \tan y dy = \log c$
or $\log \sec x \sec y = \log c$ **or** $\sec x \sec y = c$
curve passes through the point $\left(0, \frac{\pi}{4}\right)$
$\sec 0 \sec \frac{\pi}{4} = \mathrm{c} = \sqrt{2}$
Hence, the required equation of the curve is $\sec x \sec y = \sqrt{2}$

Q.13 Answer:

$0.075\mathrm{x}^2 + 2\mathrm{x} + 100$

Solution:

[Note derivative of a function is sometimes called as the marginal value, specially in economics]
Given $\frac{\mathrm{dc}}{\mathrm{dx}} = 2 + 0.15\mathrm{x}$ **Integrating we get** $c(x) = 2x + 0.15\frac{x^2}{2} + D$
when $\mathrm{x} = 0, \mathrm{c(x)} = 100 \Rightarrow \mathrm{D} = 100$
$\therefore$ **the required equation is** $c(x) = 0.075x^2 + 2x + 100$

Q.14 Answer:

$40\frac{\log 3}{\log 2}$ years

Solution:

Let the intial population be x_0 **and it is** x **in years, then the differential equation is** $\frac{\mathrm{dx}}{\mathrm{dt}} = \mathrm{kx}$, **k is a constant**
$\Rightarrow \frac{\mathrm{dx}}{\mathrm{x}} = \mathrm{kdt}$. **Integrating we get**
$\log x + kt + c$
When $\mathrm{t} = 0, \mathrm{x} = \mathrm{x}_0 \Rightarrow \mathrm{c} = \log \mathrm{x}_0$
Then from (i) $\log x = \mathrm{kt} + \log x_0$
$\Rightarrow \log \frac{\mathrm{x}}{\mathrm{x}_0} = \mathrm{kt}$
Now when
$\mathrm{t} = 40, \frac{\mathrm{x}}{\mathrm{x}_0} = 2 \Rightarrow \log 2 = \mathrm{k}.40 \Rightarrow \mathrm{k} = \frac{\log 2}{40}$
$\therefore$ **(ii) becomes** $\log \frac{\mathrm{x}}{\mathrm{x}_0} = \frac{\log 2}{40} \cdot \mathrm{t}$
Next put $\frac{\mathrm{x}}{\mathrm{x}_0} = 3 \Rightarrow \mathrm{t} = 40\frac{\log 3}{\log 2}$

Q.15 Answer:

$\mathrm{a} = 0, \mathrm{c} \neq 0, \ \mathrm{b} \neq 0$

Solution:

Given: $\frac{dy}{dx} = \frac{ax+b}{cy+d}$
or, $(\mathrm{cy} + \mathrm{d})\mathrm{dy} = (\mathrm{ax} + \mathrm{b})\mathrm{dx}$
Integrating both the sides.
c. $\int ydy + d\int dy = a\int xdx + b\int dx + K$

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[K is constant integration]

or, c. $\frac{y^2}{2} + d \cdot y = a\frac{x^2}{2} + b \cdot x + K$

or, $cy^2 + 2d \cdot y = ax^2 + 2b \cdot x + 2K$

This equation will represent a parabola when either, the coefficient of x^2 or the coefficient of y^2 is zero, but not both. Thus either $c = 0$ or $a = 0$ but not both . From the choice given, $a = 0, c \neq 0$ and $b \neq 0$.

Q.16 Answer:

$y^2 = x^2 + 2xy\frac{dy}{dx}$

Solution:

Let the equation be $x^2 + y^2 + 2gx = 0 \ldots$ (i)

Differentiating (i) w.r.t x and eliminating g, we get

$y^2 = x^2 + 2xy\frac{dy}{dx}$

Q.17 Answer:

$y = x\cot^{-1}(\log_e ex)$

Solution:

Givea $\frac{dy}{dx} = \frac{y}{x} - \sin^2\left(\frac{y}{x}\right)$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$v + x\frac{dv}{dx} = v - \sin^2 v \quad \Rightarrow -\operatorname{cosec}^2 \psi d\nu = \frac{dx}{x}$

Integrating both sides,

$-\int \operatorname{cosec}^2 x dy = \int \frac{dx}{x} \quad \Rightarrow \cot v = \log x + c, \quad \cot\frac{y}{x} = \log x + c$

This curve passes through the point $\left(1, \frac{\pi}{4}\right)$

$\therefore c = 1 \Rightarrow \cot\frac{y}{x} = \log x + \log_e e$

$\cot\frac{y}{x} = \log xe \Rightarrow y = x\cot^{-1}(\log xe)$

Q.18 Answer:

hyperbolas of variable eccentricity

Solution:

Given $\frac{xdx}{1+x^2} = \frac{ydy}{1+y^2}$

Integrating we get, $\frac{1}{2}\log\left(1 + x^2\right) = \frac{1}{2}\log\left(1 + y^2\right) + a$

$\Rightarrow 1 + x^2 = c\left(1 + y^2\right), c = e^{2a}$

$x^2 - cy^2 = c - 1 \Rightarrow \frac{x^2}{c-1} - \frac{y^2}{\left(\frac{c-1}{c}\right)} = 1$

Clearly $c > 0$ as $c = e^{2a}$ Hence, the equation (i) gives a family of hyperbolas with eccentricity

$= \sqrt{\frac{c-1+\frac{c-1}{c}}{c-1}} = \sqrt{\frac{c^2-1}{c-1}} = \sqrt{c+1}$ if $c \neq 1$

Thus eccenticity varies from member to member of the family as it depends on c. If $c = 1$, it is a pair of lines $x^2 - y^2 = 0$

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Q.19 Answer:

$x-\frac{y^2}{2}+\frac{1}{3}\left(x^2+y^2\right)^{3/2}=C$

Solution:

Rearranging the equation, we have $dx-ydy+\sqrt{(x^2+y^2)}(xdx+ydy)=0$

$\Rightarrow \quad dx-ydy+\frac{1}{2}\sqrt{(x^2+y^2)}d\left(x^2+y^2\right)=0$

On integrating, we get $x-\frac{y^2}{2}+\frac{1}{2}\int\sqrt{t}dt=c,\left\{t=\sqrt{(x^2+y^2)}\right\}$

or $x-\frac{y^2}{2}+\frac{1}{3}\left(x^2+y^2\right)^{3/2}=c$

Q.20 Answer:

$a^2\frac{d^2y}{dx^2}+x^2\frac{dy}{dx}+y=0$

Solution:

From the question,

$a^2\frac{d^2y}{dx^2}+x^2\frac{dy}{dx}+y=e^x$

Since, $\mathrm{f(x)}$ **and** $\mathrm{g(x)}$ **are two solutions,**

$a^2f''(x)+x^2f'(x)+f(x)=e^x$ **and** $a^2f''(x)+x^2f'(x)+f(x)=e^x$

By subtracting, we get

$a^2\left[f''(x)-g''(x)\right]+\left[f(x)-g'(x)\right]+\left[f(x)-g(x)\right]=0$

Hence, $\mathrm{f(x)-g(x)}$ **is the solution is**

$a^2\frac{d^2y}{dx^2}+x^2\frac{dy}{dx}+y=0$

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